THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHEMATICS UNIT TEST SET A

Chapters 8–11: **Introduction to Trigonometry, Some Applications of Trigonometry, Circles, Areas Related to Circles** | RBSE Board Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–12 are Objective Type (1 mark each).
3. Q. 13–17 are Very Short Answer Type (2 marks each).
4. Q. 18–23 are Short Answer Type (3 marks each).
5. Q. 24–25 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 22 to 25.
7. Write the question number before each answer.
8. Draw neat figures; use graph paper where a graph is asked.

SECTION — A (Objective Type Questions) — 12 × 1 = 12

Choose the correct option and write it in your answer book.

1. If $2\sin\theta = 1$ (θ acute), then θ is: 1

(a) 60° **(b)** 30° **(c)** 90° **(d)** 45°

2. If $\tan A = \frac{4}{3}$, then $\sin A$ is: 1

(a) $\frac{3}{4}$ **(b)** $\frac{3}{5}$ **(c)** $\frac{5}{4}$ **(d)** $\frac{4}{5}$

3. The value of $\sin 30^\circ + \cos 60^\circ$ is: 1

(a) $\frac{\sqrt{3}}{2}$ **(b)** 0 **(c)** $\frac{1}{2}$ **(d)** 1

4. A kite string is 100 m long and makes 30° with the ground. The height of the kite is: 1

(a) 50 m **(b)** $\frac{100}{\sqrt{3}}$ m **(c)** 100 m **(d)** $50\sqrt{3}$ m

5. The angle of elevation of the top of a tower from a point 100 m away is 60°. Its height is: 1

(a) 50 m **(b)** $100\sqrt{3}$ m **(c)** 200 m **(d)** $\frac{100}{\sqrt{3}}$ m

6. A ladder leaning against a wall makes 60° with the ground. Its foot is 2.5 m from the wall. The length of the ladder is: 1

(a) $5\sqrt{3}$ m **(b)** $2.5\sqrt{3}$ m **(c)** 2.5 m **(d)** 5 m

7. PQ is a tangent at P to a circle with centre O. $OP = 5$ cm and $OQ = 13$ cm. Student A says $PQ = 12$ cm; Student B says $PQ = 8$ cm ($= 13 - 5$). Then: 1

(a) Student B is correct **(b)** Student A is correct

(c) Both are correct **(d)** Both are wrong

8. The common point of a tangent and a circle is called the: 1

(a) chord **(b)** centre **(c)** end point **(d)** point of contact

9. A quadrilateral ABCD circumscribes a circle. If $AB = 6$ cm, $BC = 7$ cm and $CD = 4$ cm, then AD is: 1

(a) 3 cm **(b)** 5 cm **(c)** 9 cm **(d)** 4 cm

10. The length of an arc of a circle of radius 14 cm subtending 30° at the centre is: 1

(a) $\frac{11}{3}$ cm **(b)** 22 cm **(c)** $\frac{22}{3}$ cm **(d)** $\frac{44}{3}$ cm

11. The area of a quadrant of a circle whose circumference is 22 cm is: 1

(a) $\frac{77}{2}$ cm² **(b)** $\frac{77}{8}$ cm² **(c)** 154 cm² **(d)** $\frac{77}{4}$ cm²

12. The area of a semicircle of radius 7 cm is: 1

(a) 154 cm² **(b)** 38.5 cm² **(c)** 77 cm² **(d)** 22 cm²

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

13. Evaluate: $\frac{5\cos^2 60^\circ + 4\sec^2 30^\circ - \tan^2 45^\circ}{\sin^2 30^\circ + \cos^2 30^\circ}$. 2

14. Given $\sec\theta = \frac{13}{12}$, calculate all the other trigonometric ratios. **2**

15. An electrician has to repair a fault on a 5 m high pole at a point 1.3 m below its top. His ladder is inclined at 60° to the horizontal. Find the length of the ladder needed. **2**

16. Two concentric circles have radii 10 cm and 6 cm. Find the length of the chord of the larger circle which touches the smaller circle. **2**

17. An umbrella has 8 ribs which are equally spaced. Assuming it to be a flat circle of radius 45 cm, find the area between two consecutive ribs. **2**

SECTION — C (Short Answer Type) — 6 × 3 = 18

Answer in about 40–60 words each.

18. If $3\cot A = 4$, check whether $\frac{1-\tan^2 A}{1+\tan^2 A} = \cos^2 A - \sin^2 A$. **3**

19. The shadow of a tower standing on level ground is found to be 40 m longer when the sun's altitude is 30° than when it is 60°. Find the height of the tower. **3**

20. Prove that the opposite sides of a quadrilateral circumscribing a circle subtend supplementary angles at the centre of the circle. **3**

21. To warn ships of underwater rocks, a lighthouse spreads a red light over a sector of angle 80° to a distance of 16.5 km. Find the area of the sea over which ships are warned. (Use $\pi = 3.14$) **3**

22. Prove that $(\csc\theta - \cot\theta)^2 = \frac{1-\cos\theta}{1+\cos\theta}$. **3**

OR

Prove that $\frac{1+\sec A}{\sec A} = \frac{\sin^2 A}{1-\cos A}$. **3**

23. Two tangents TP and TQ are drawn to a circle with centre O from an external point T. Prove that $\angle PTQ = 2\angle OPQ$. **3**

OR

Prove that a parallelogram circumscribing a circle is a rhombus. **3**

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

24. A man on the deck of a ship is 10 m above water level. He observes that the angle of elevation of the top of a hill is 60° and the angle of depression of the base of the hill is 30°. Find the distance of the hill from the ship and the height of the hill. **5**

OR

From the top of a 60 m high building, the angles of depression of the top and the bottom of a tower are 45° and 60° respectively. **5**

(i) Find the distance between the building and the tower.

(ii) Find the height of the tower.

(iii) A student says the tower is taller than the building. Comment.

25. In a circle of radius 14 cm, a chord PQ subtends 120° at the centre. Find **5**

(i) the length of arc PQ

(ii) the area of the minor sector

(iii) the area of the minor segment

(iv) the area of the major segment. (Use $\sqrt{3} = 1.73$)

OR

Find the area of the minor segment of a circle of radius 14 cm if the length of the corresponding arc is 22 cm. Also find the area of the major segment. **5**

END OF QUESTION PAPER — **Total: 50 Marks**